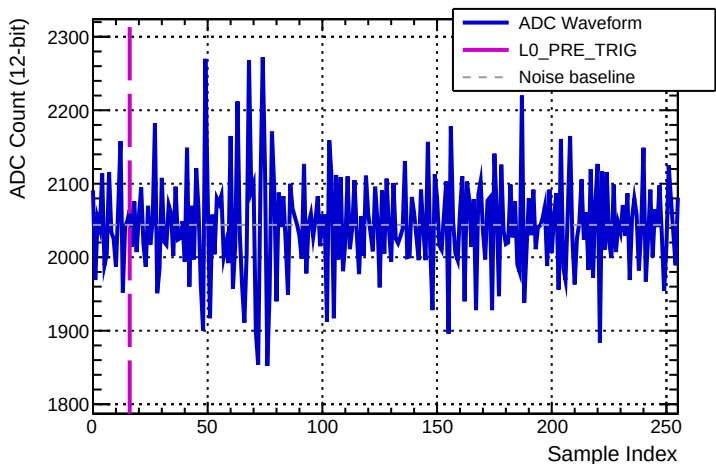
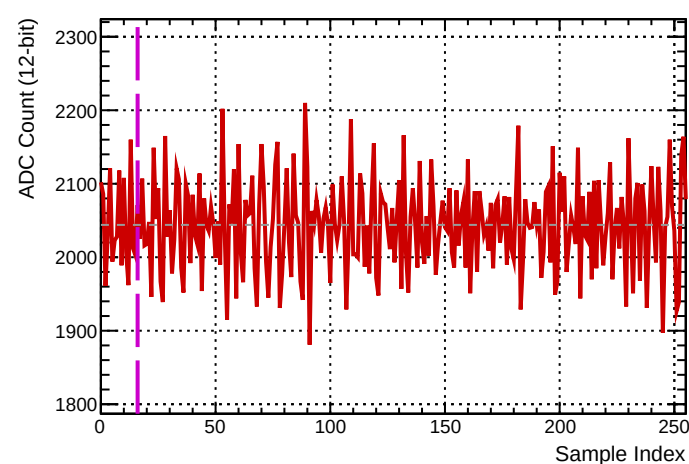
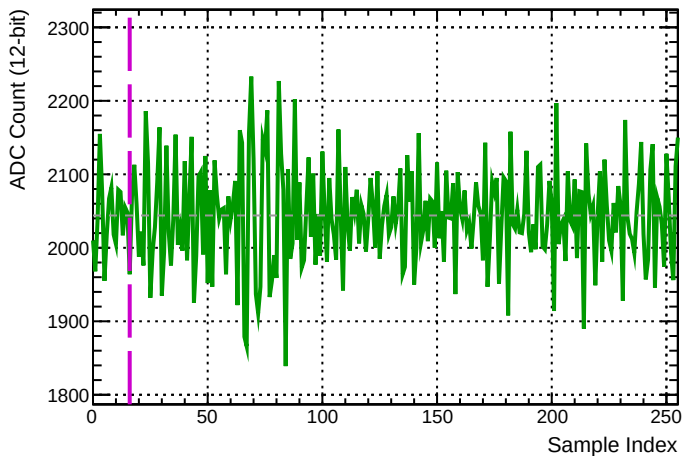
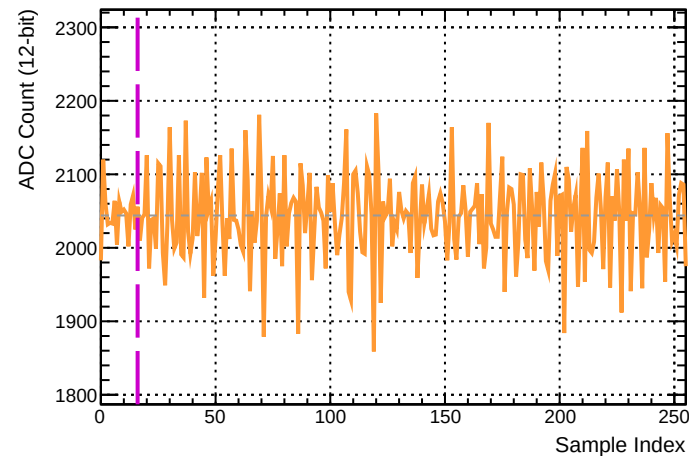


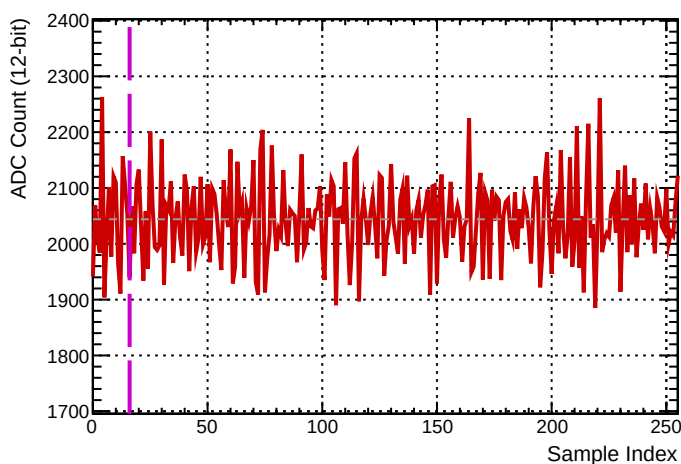
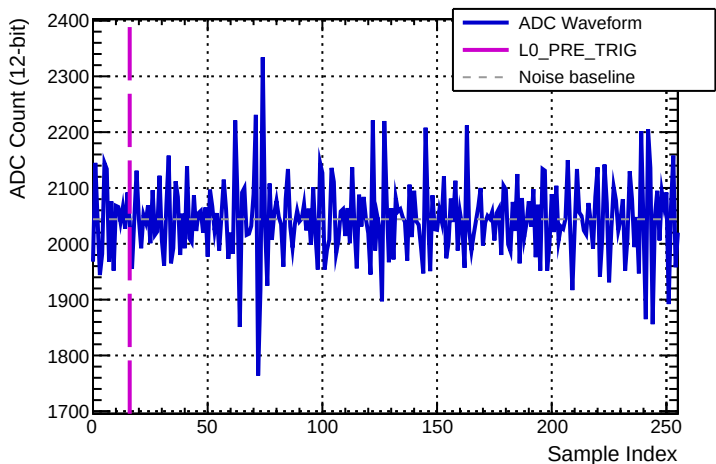
AI Trigger System -- Simulation Results

20 injected events | 21 triggers | 20 CNN outputs

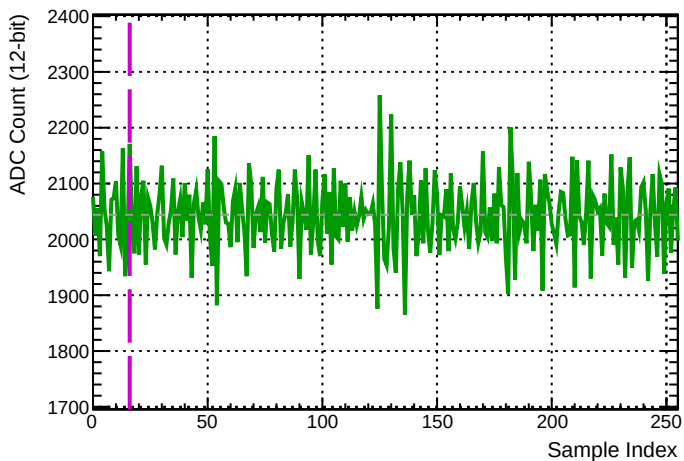
62.5 MHz fast clock | 200 MHz CNN clock | 12-bit ADC

Ch 0 | Event #9 | Label: **SIGNAL** | CNN Score: 128137Ch 1 | Event #9 | Label: **SIGNAL** | CNN Score: 128137Ch 2 | Event #9 | Label: **SIGNAL** | CNN Score: 128137Ch 3 | Event #9 | Label: **SIGNAL** | CNN Score: 128137

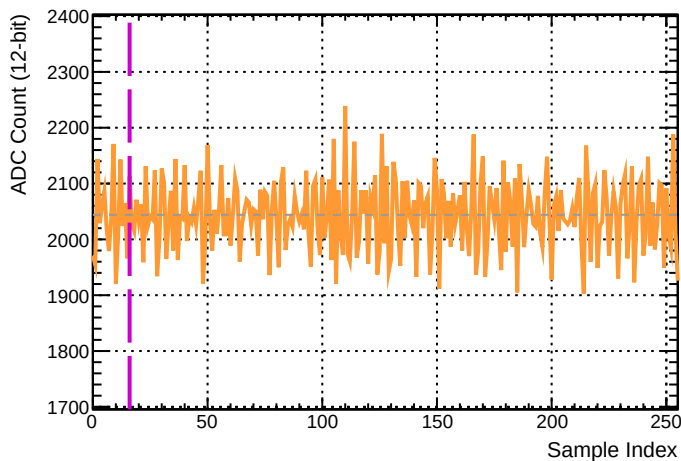
Ch 0 | Event #709 | Label: Background | Sim Test Dataset Event #709 | Label: Background | CNN Score: 7941

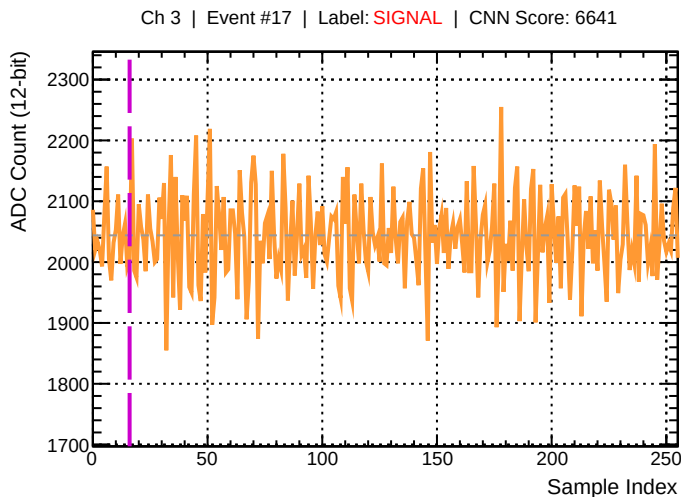
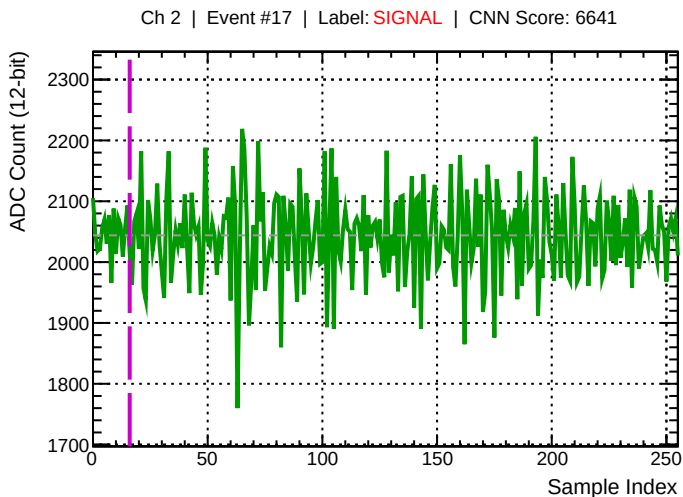
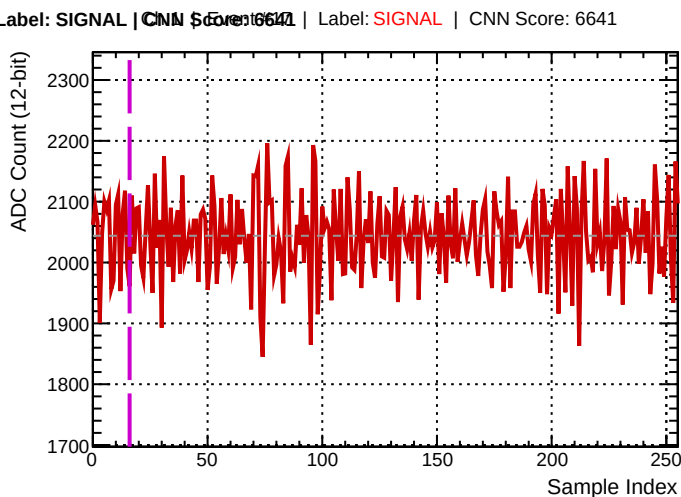
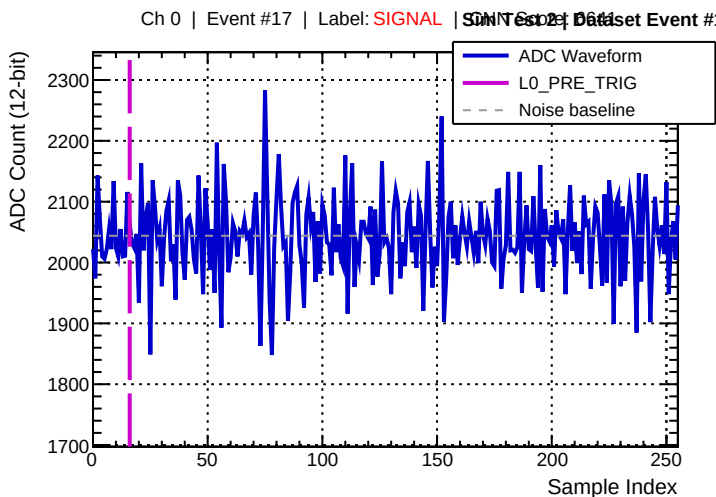


Ch 2 | Event #709 | Label: Background | CNN Score: 7941

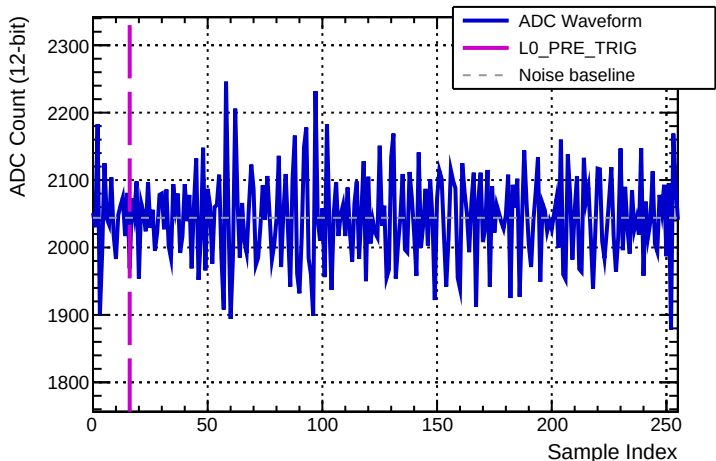


Ch 3 | Event #709 | Label: Background | CNN Score: 7941

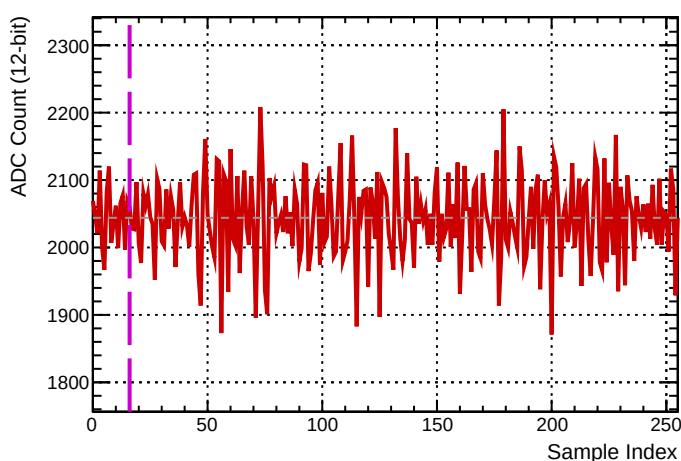




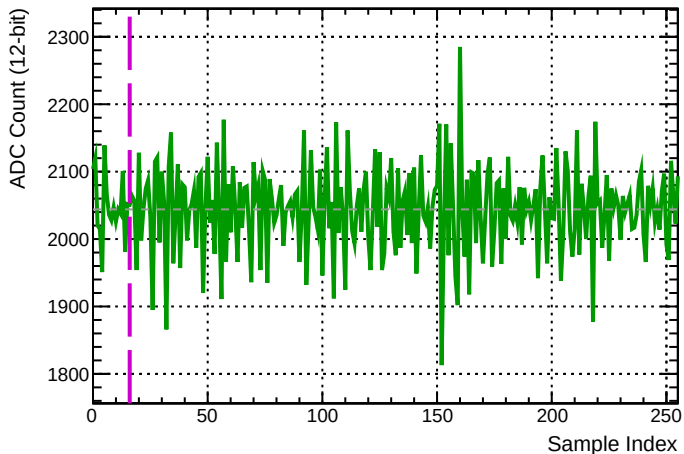
Ch 0 | Event #913 | Label: Background | CNN Score: 129505



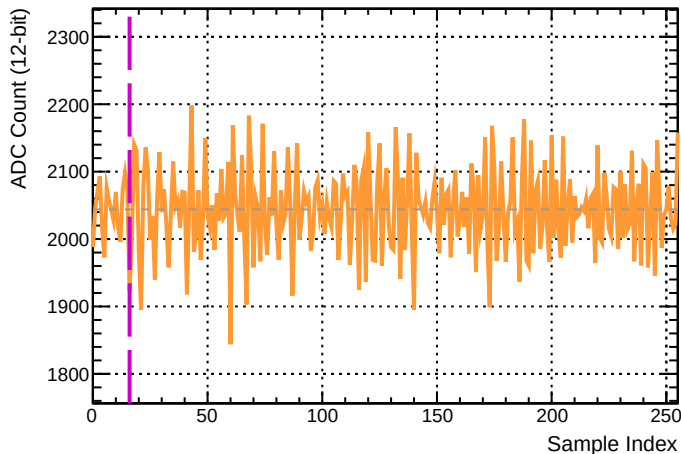
Ch 1 | Event #913 | Label: Background | CNN Score: 129505

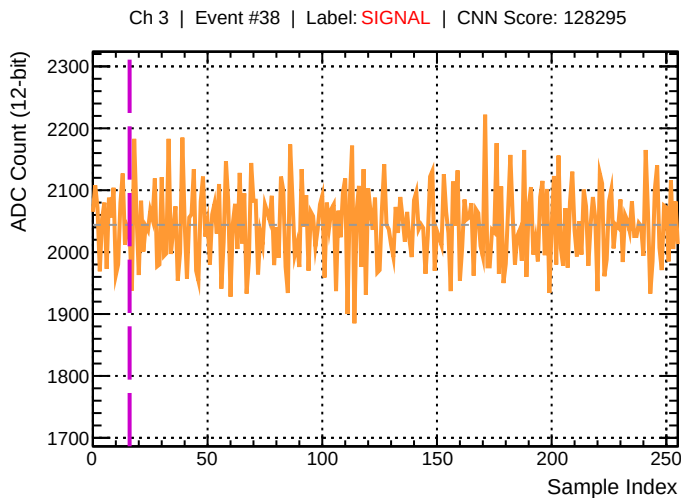
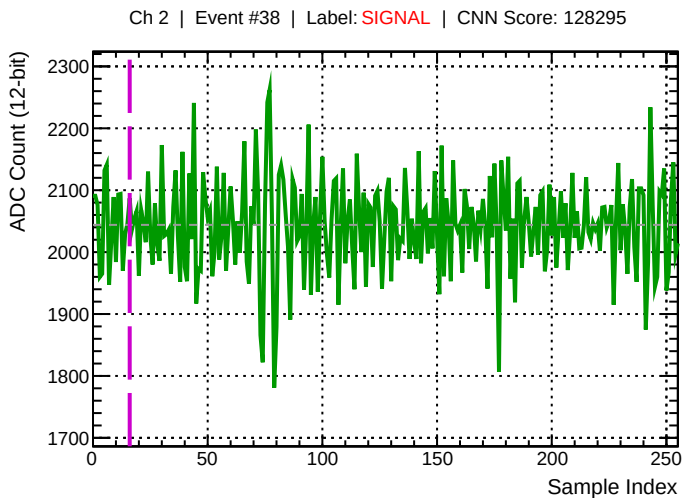
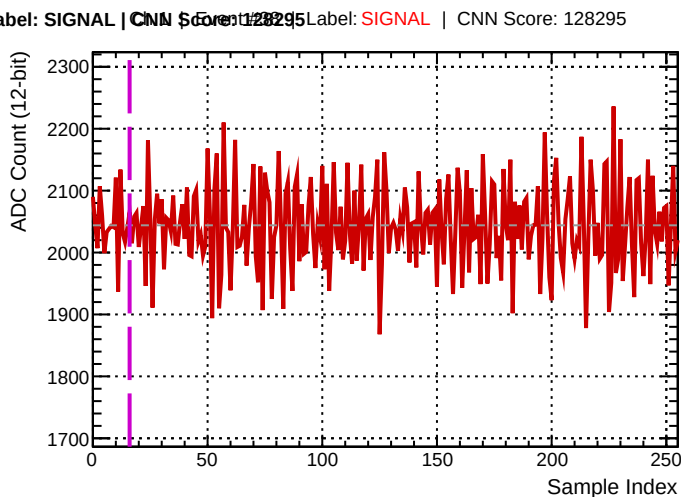
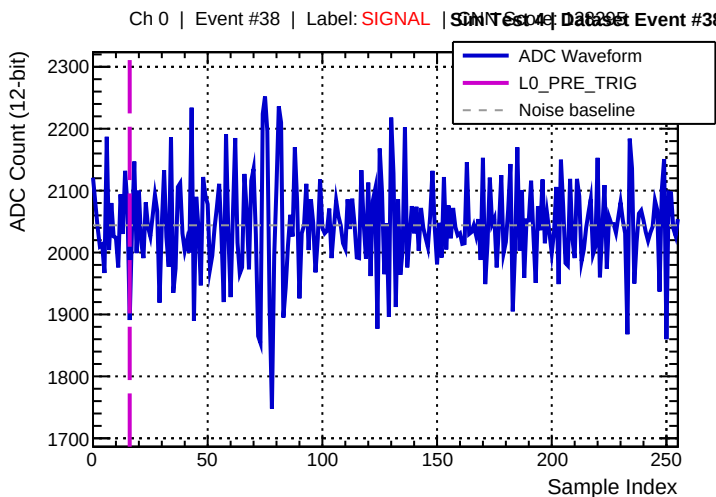


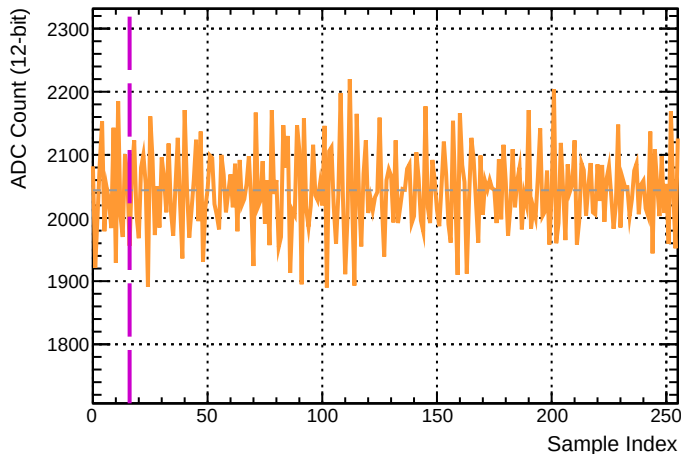
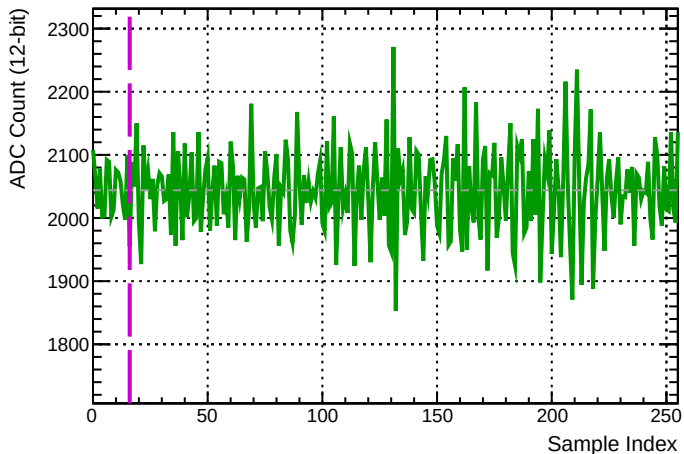
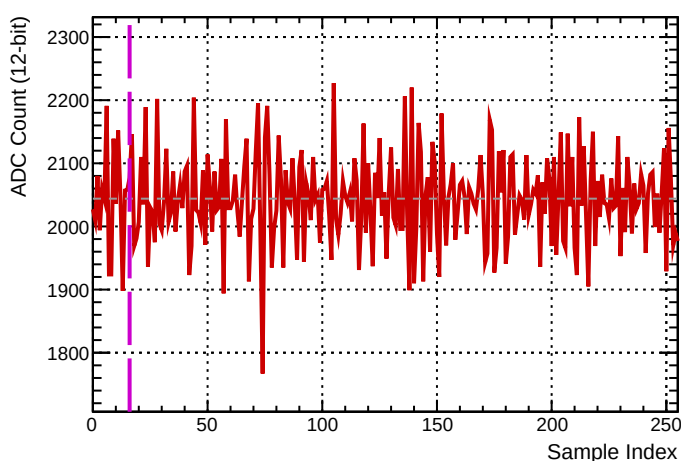
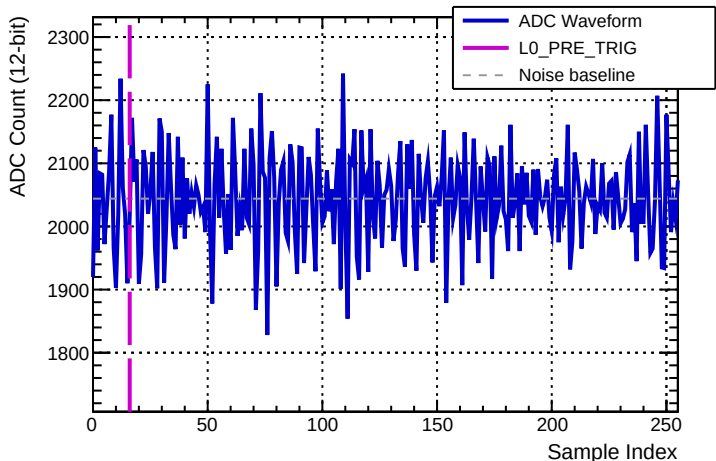
Ch 2 | Event #913 | Label: Background | CNN Score: 129505

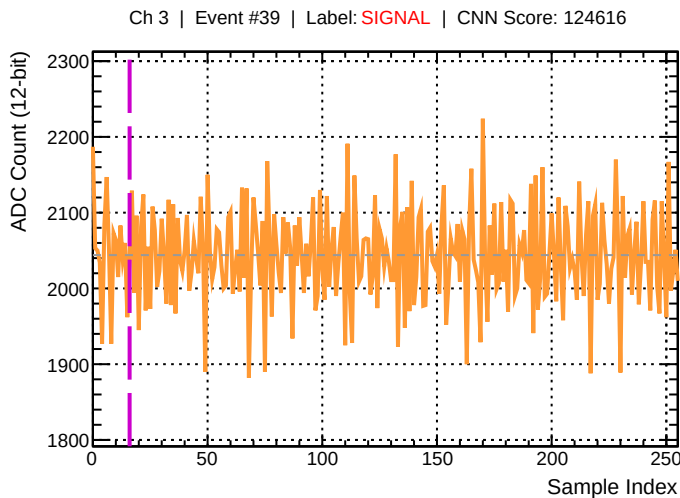
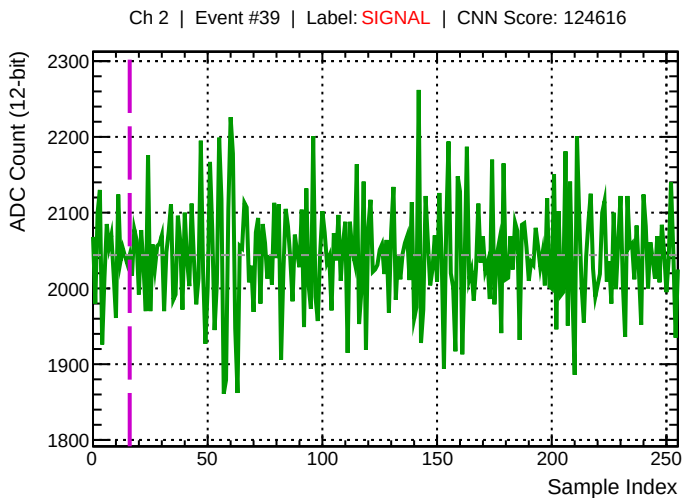
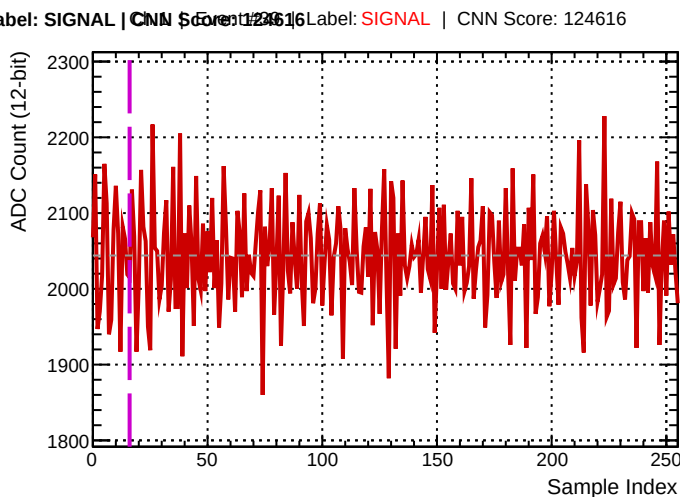
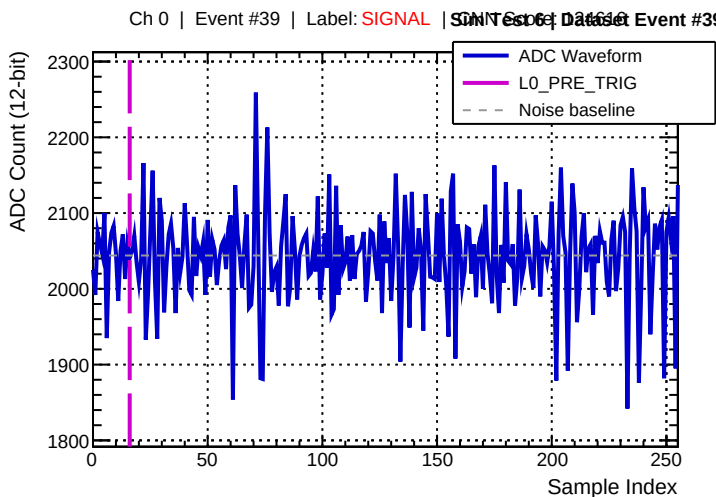


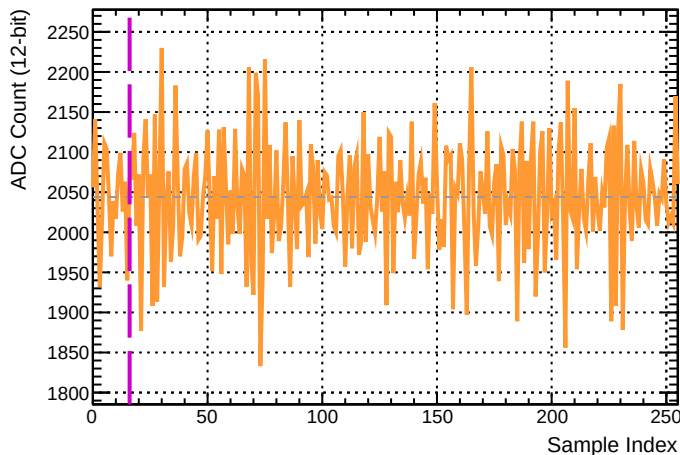
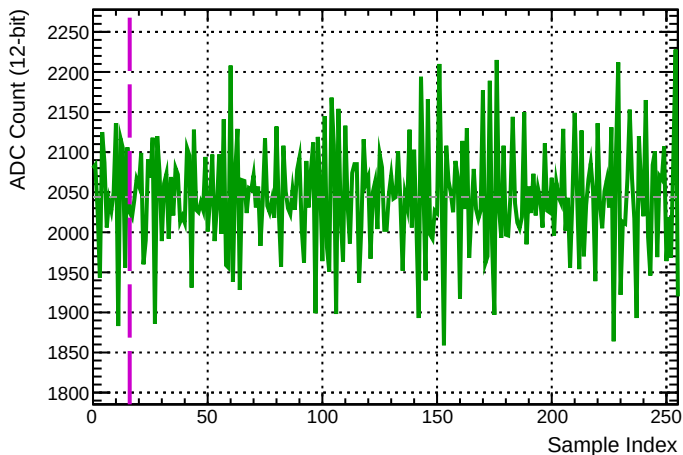
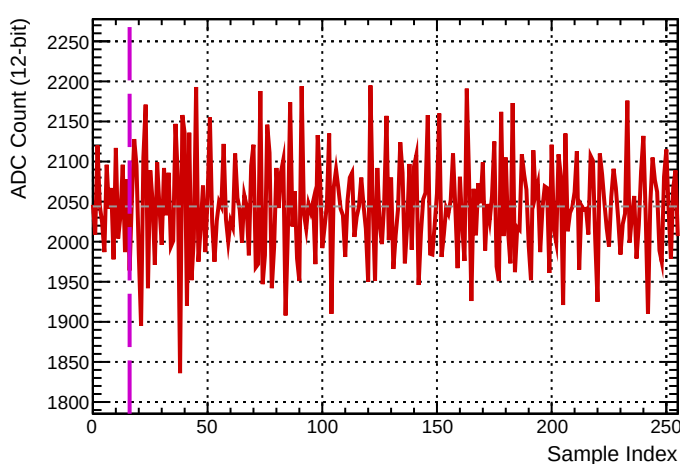
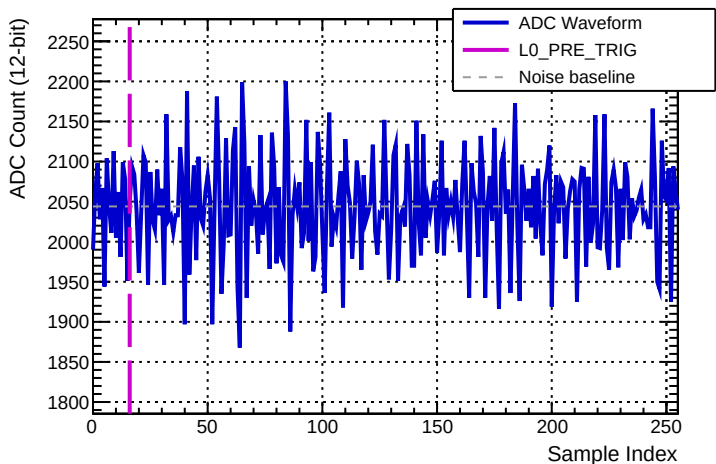
Ch 3 | Event #913 | Label: Background | CNN Score: 129505

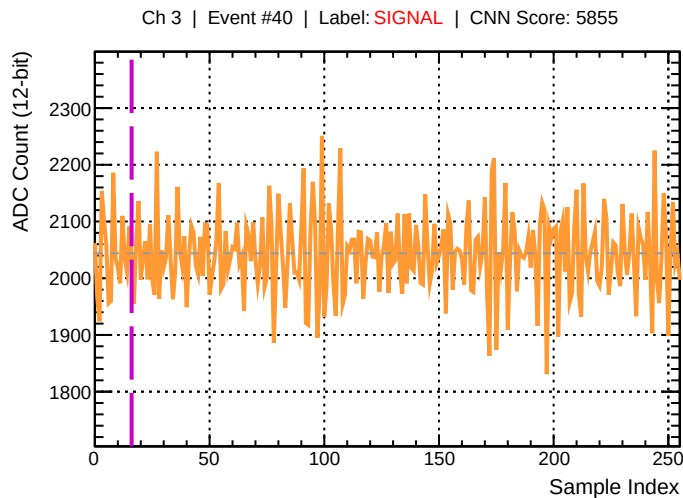
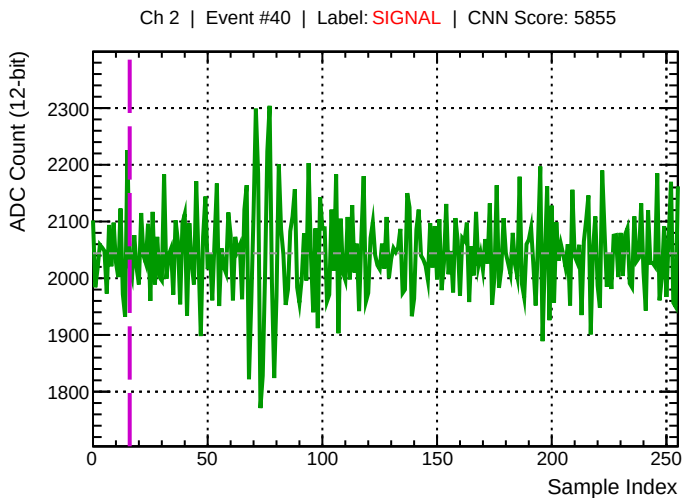
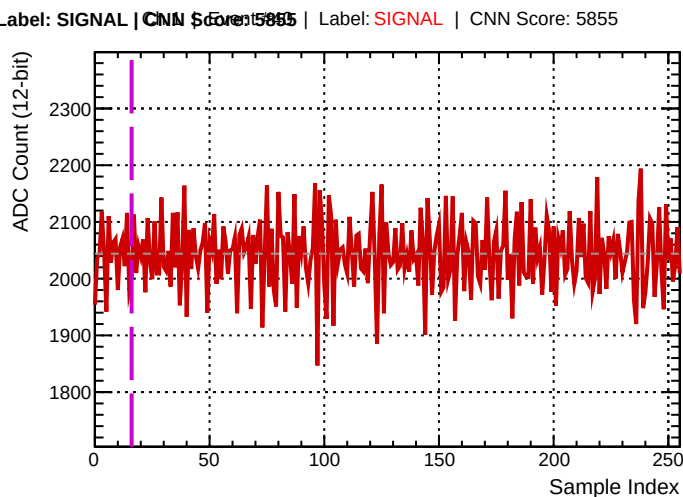
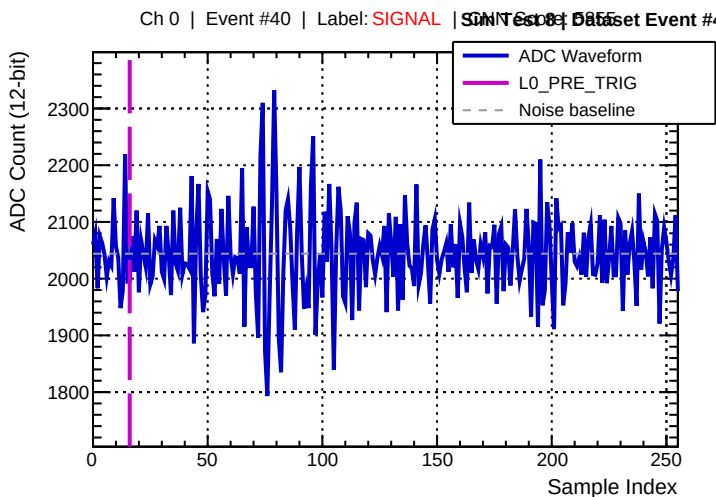




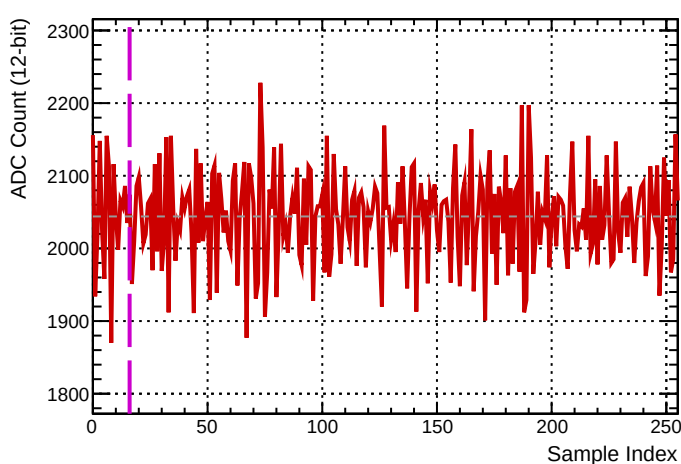
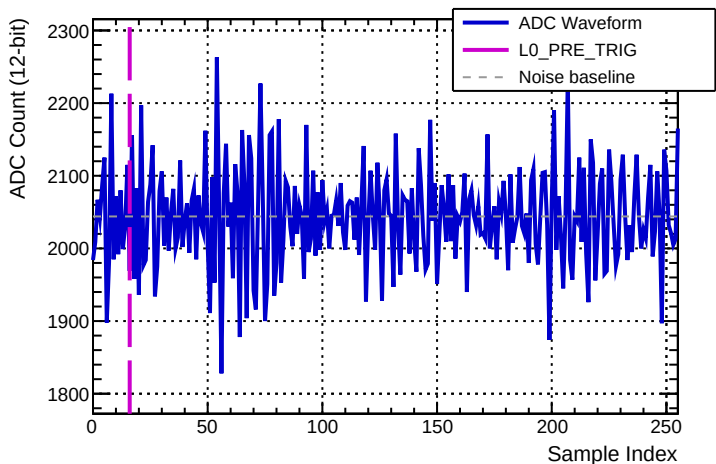




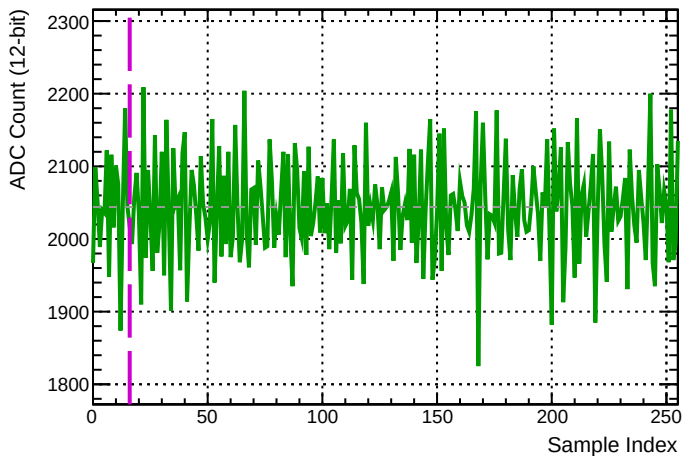




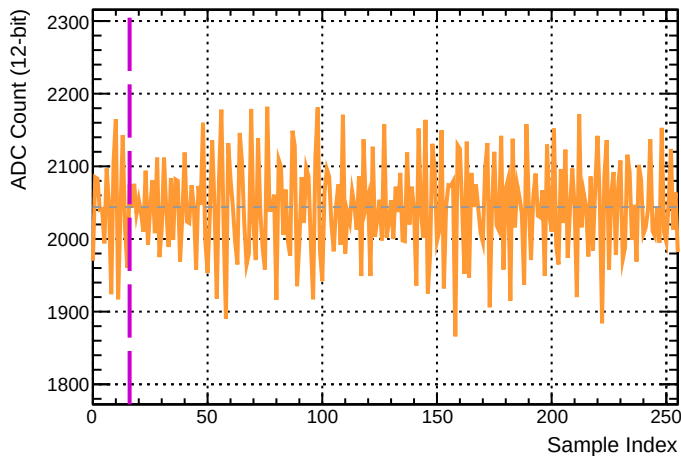
Ch 0 | Event #236 | Label: Background | CNN Score: 3642



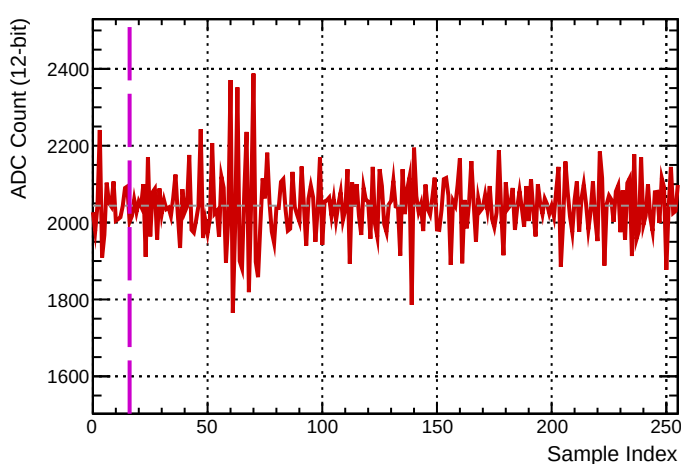
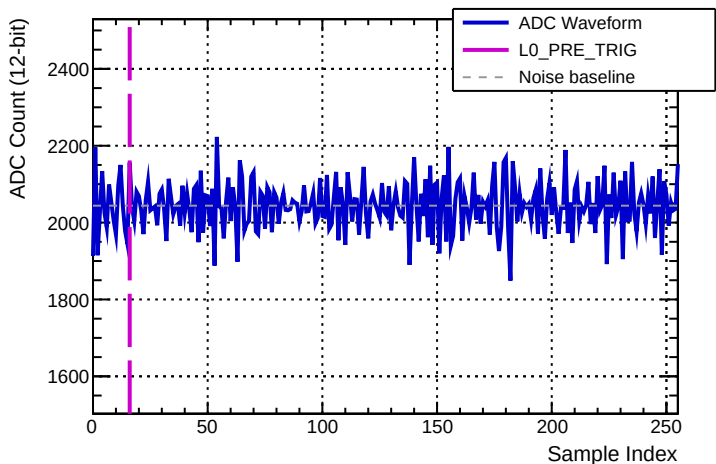
Ch 2 | Event #236 | Label: Background | CNN Score: 3642



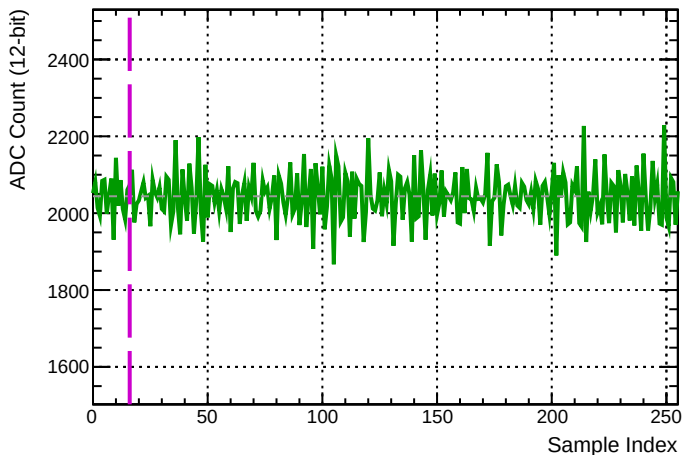
Ch 3 | Event #236 | Label: Background | CNN Score: 3642



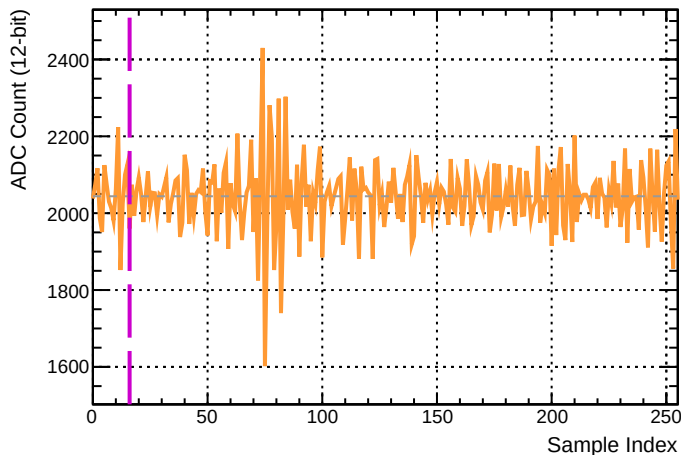
Ch 0 | Event #41 | Label: **SIGNAL** | CNN Score: 3763 | Dataset Event #41 | Label: **SIGNAL** | CNN Score: 3763



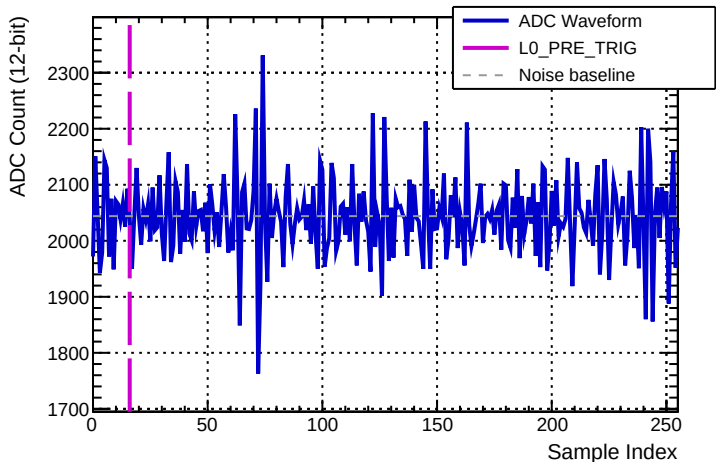
Ch 2 | Event #41 | Label: **SIGNAL** | CNN Score: 3763



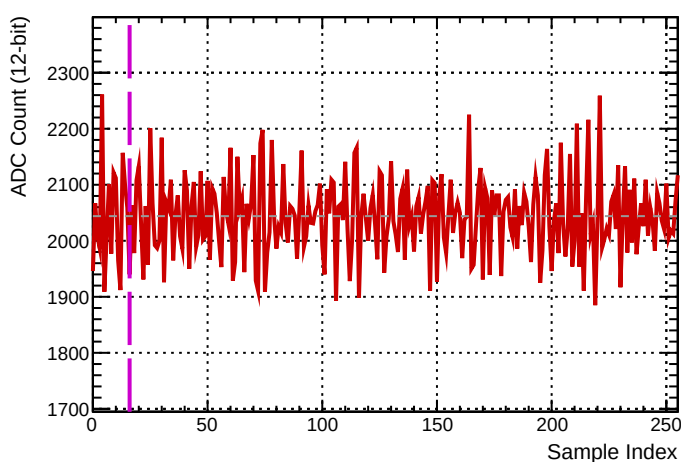
Ch 3 | Event #41 | Label: **SIGNAL** | CNN Score: 3763



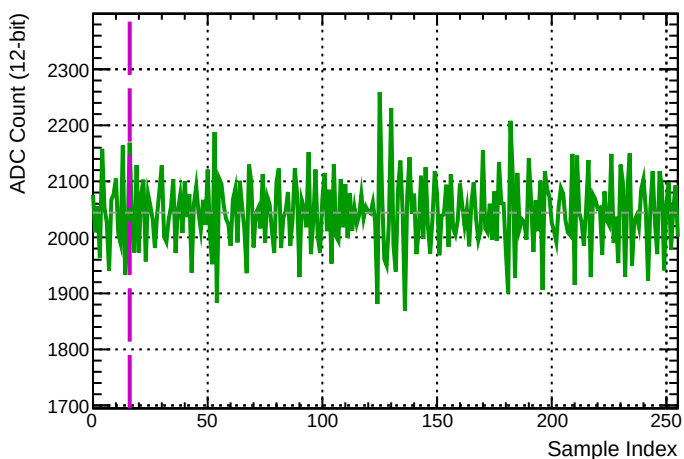
Ch 0 | Event #709 | Label: **Background** | CNN Score: 123878



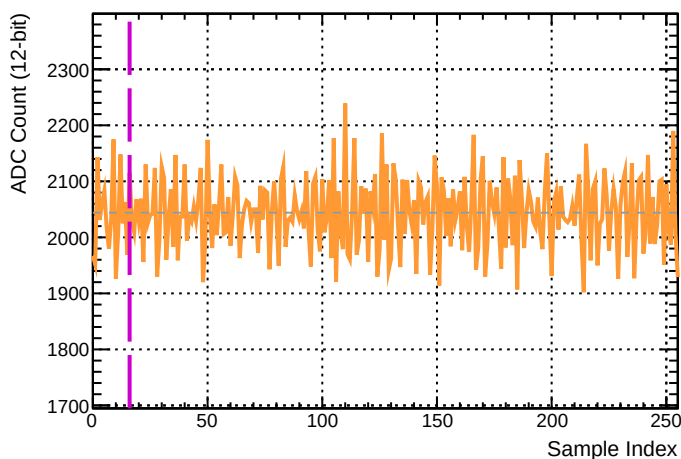
Ch 1 | Event #709 | Label: **Background** | CNN Score: 123878



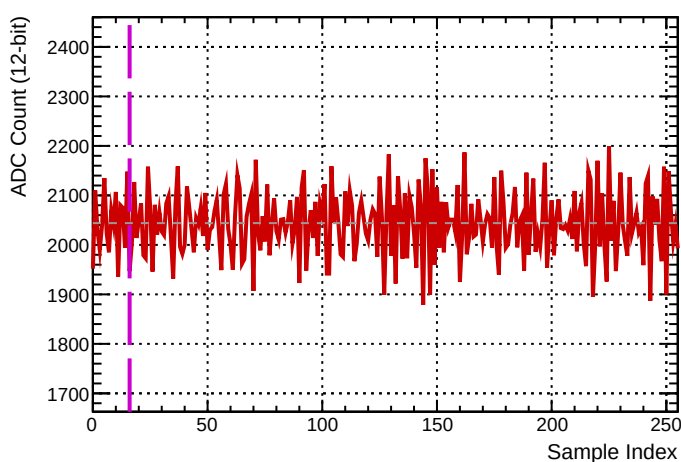
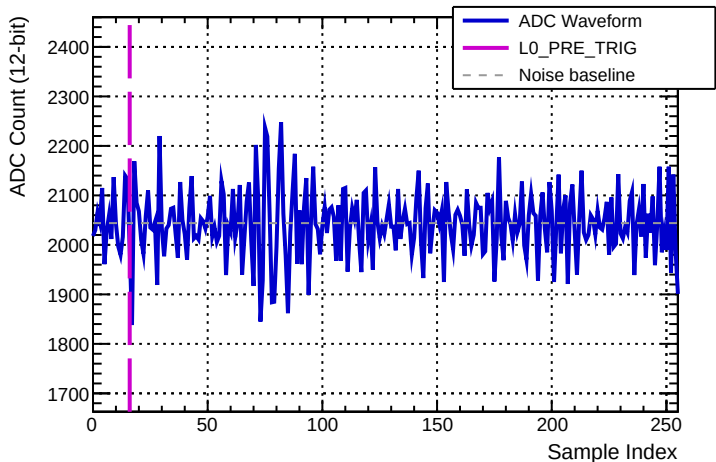
Ch 2 | Event #709 | Label: **Background** | CNN Score: 123878



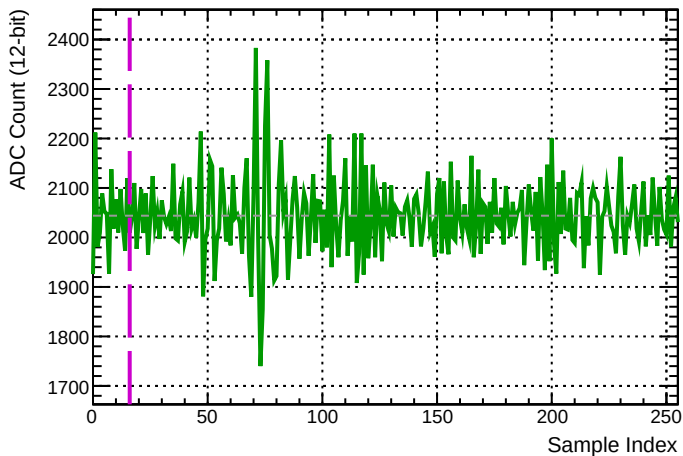
Ch 3 | Event #709 | Label: **Background** | CNN Score: 123878



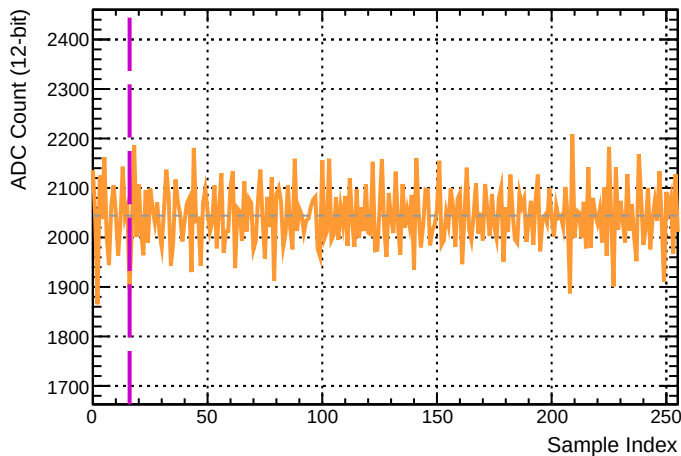
Ch 0 | Event #42 | Label: **SIGNAL** | CNN Score: 1008 | Dataset Event #42 | Label: **SIGNAL** | CNN Score: 1008

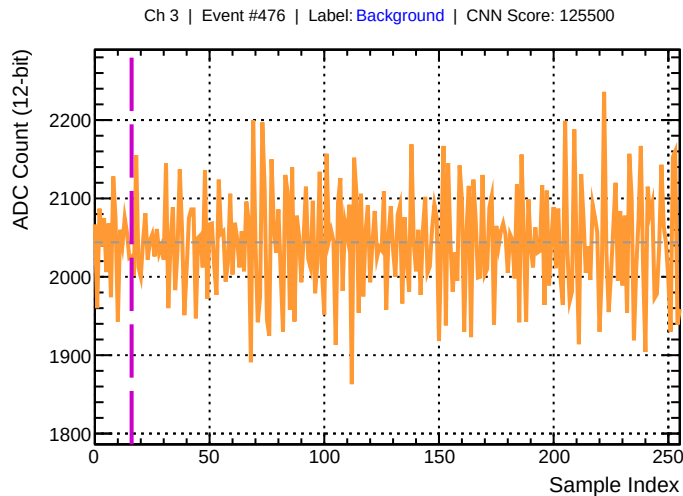
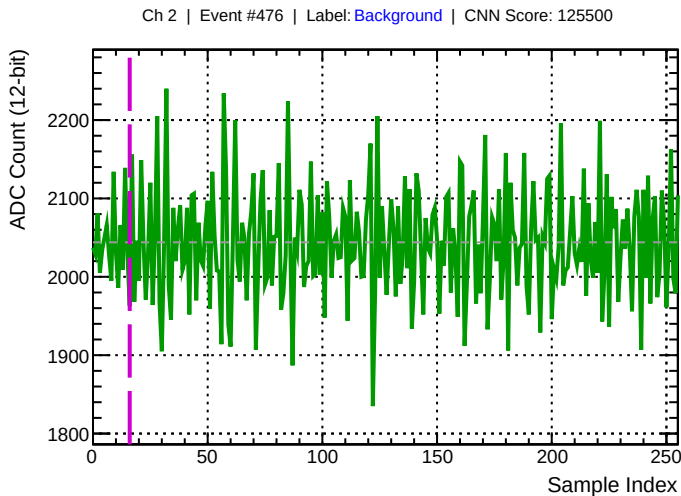
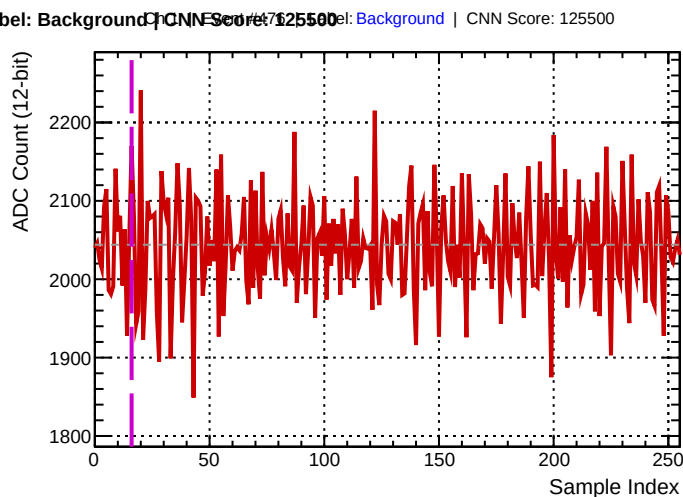
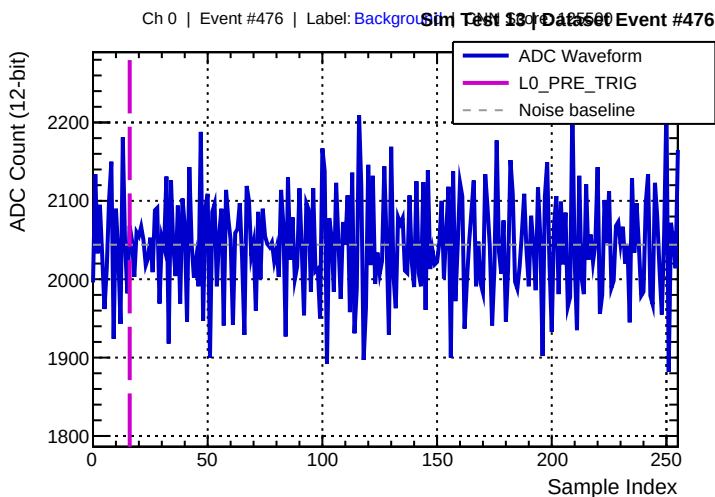


Ch 2 | Event #42 | Label: **SIGNAL** | CNN Score: 1008

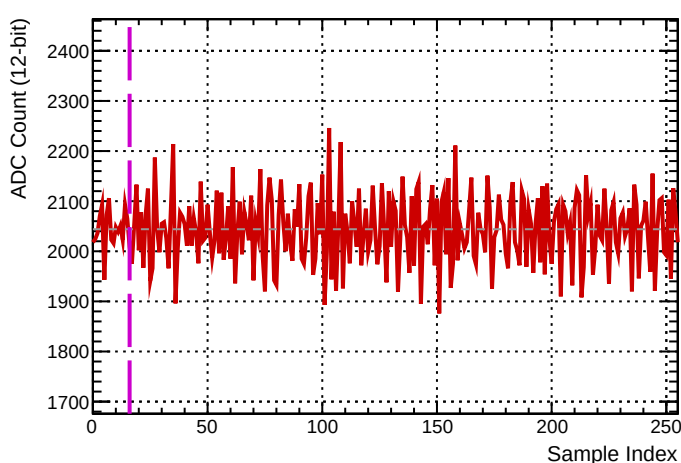
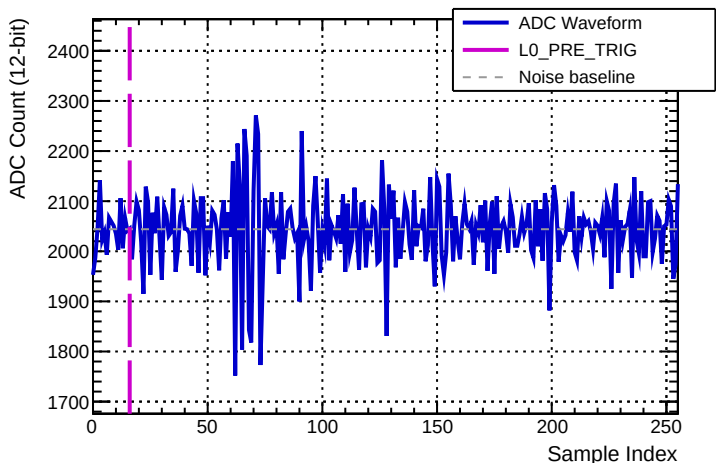


Ch 3 | Event #42 | Label: **SIGNAL** | CNN Score: 1008

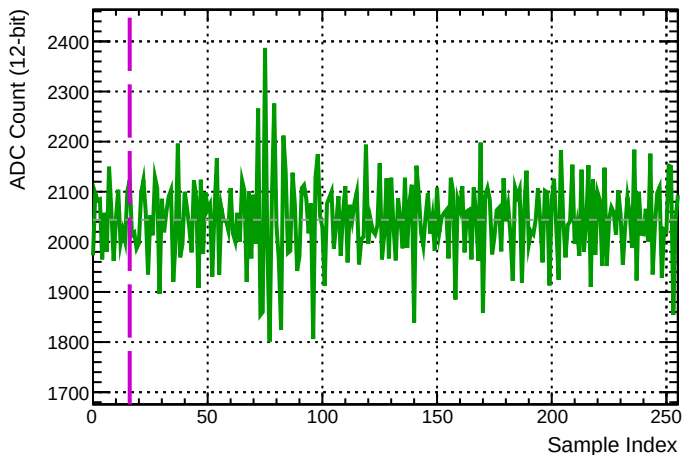




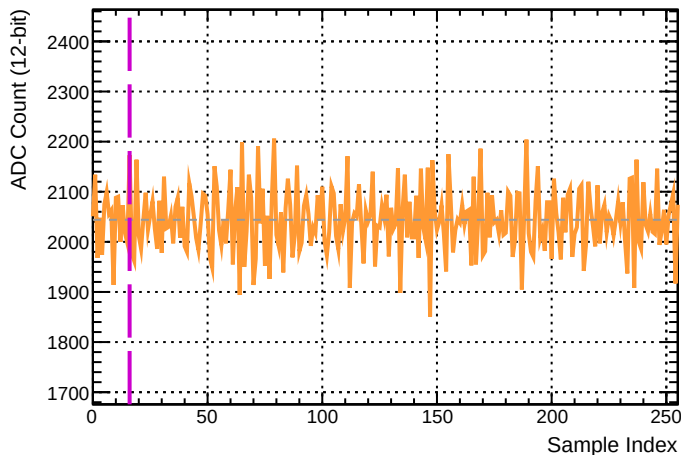
Ch 0 | Event #45 | Label: **SIGNAL** | CNN Score: 4132



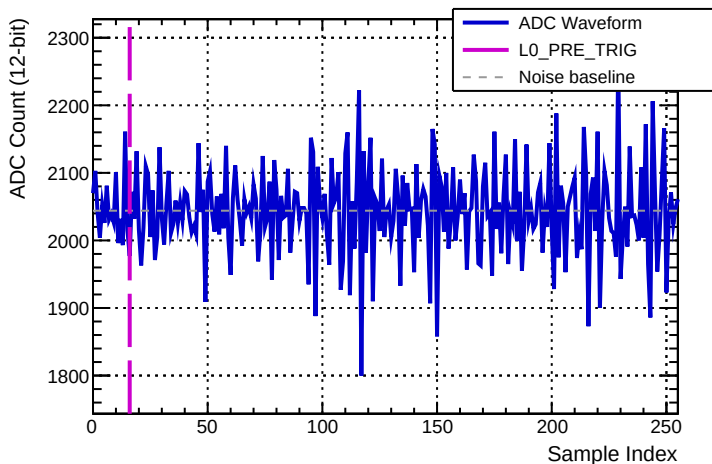
Ch 2 | Event #45 | Label: **SIGNAL** | CNN Score: 4132



Ch 3 | Event #45 | Label: **SIGNAL** | CNN Score: 4132

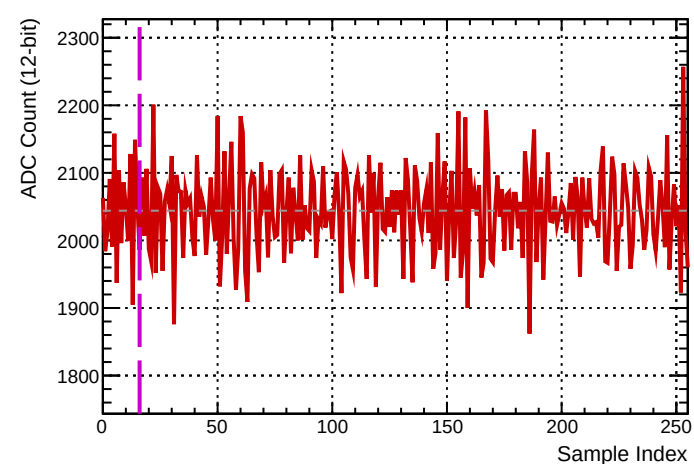


Ch 0 | Event #407 | Label: Background | CNN Score: 5376

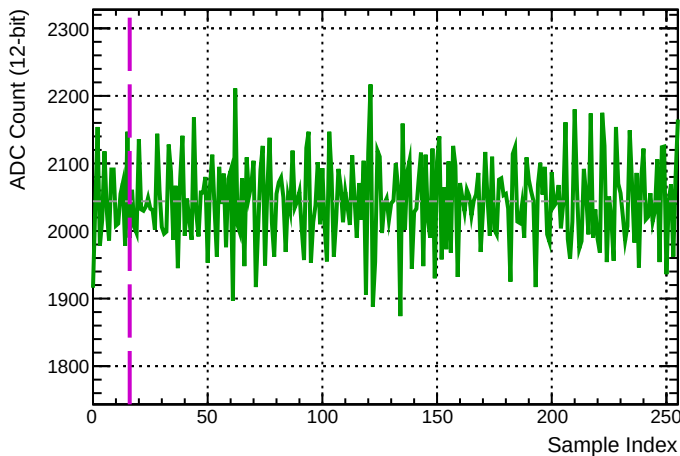


Sim Test 15 | Dataset 1

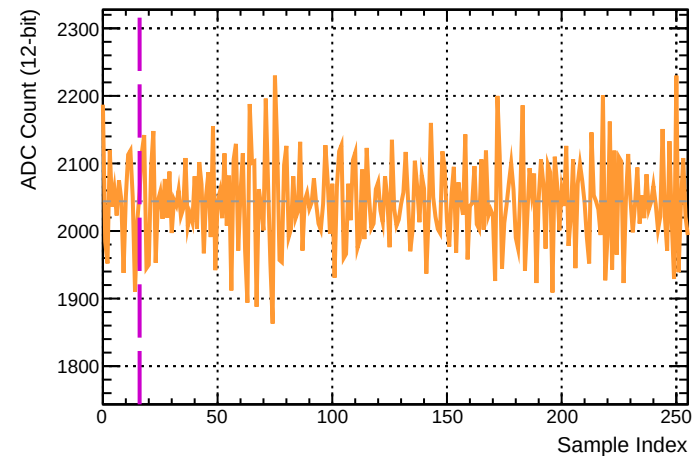
Event #407 | Label: Background | CNN Score: 5376



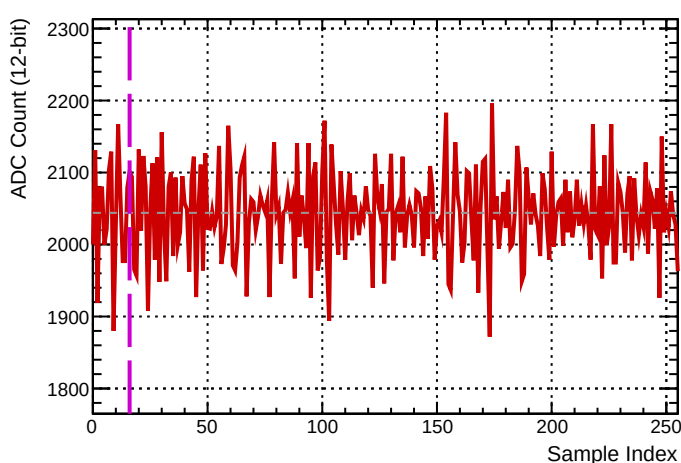
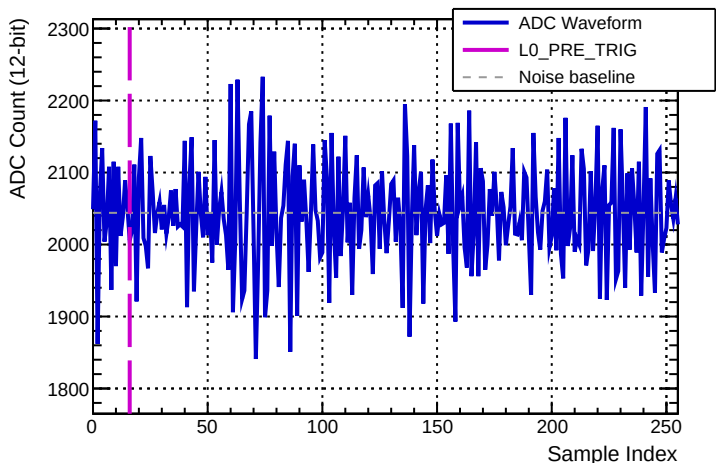
Ch 2 | Event #407 | Label: Background | CNN Score: 5376



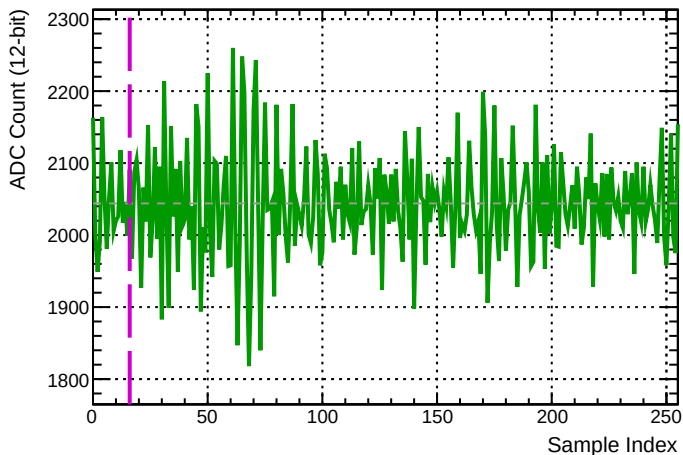
Ch 3 | Event #407 | Label: Background | CNN Score: 5376



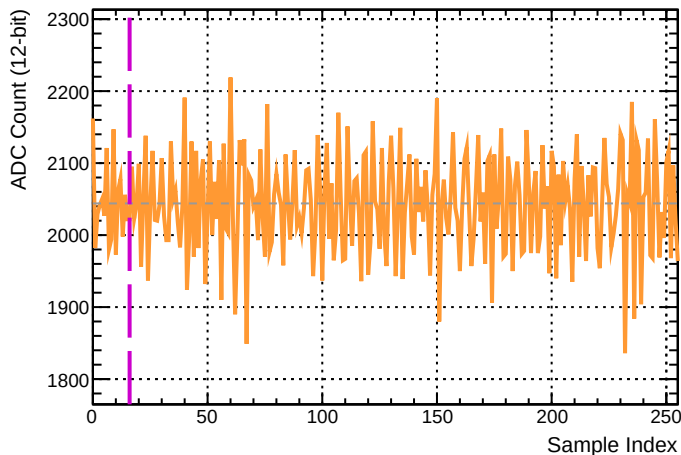
Ch 0 | Event #48 | Label: **SIGNAL** | CNN Score: 3500



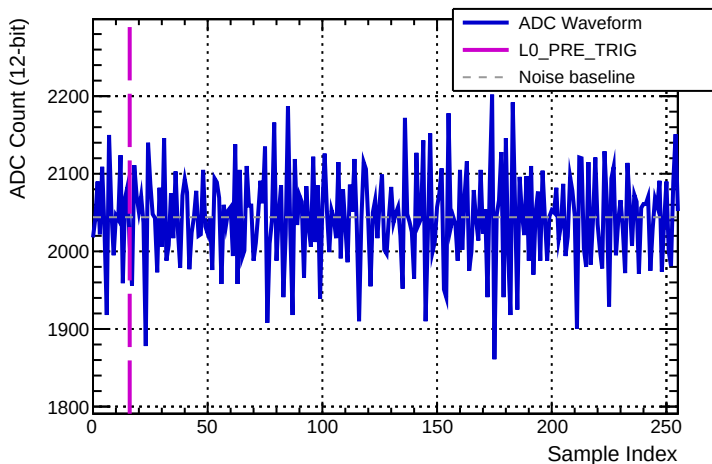
Ch 2 | Event #48 | Label: **SIGNAL** | CNN Score: 3500



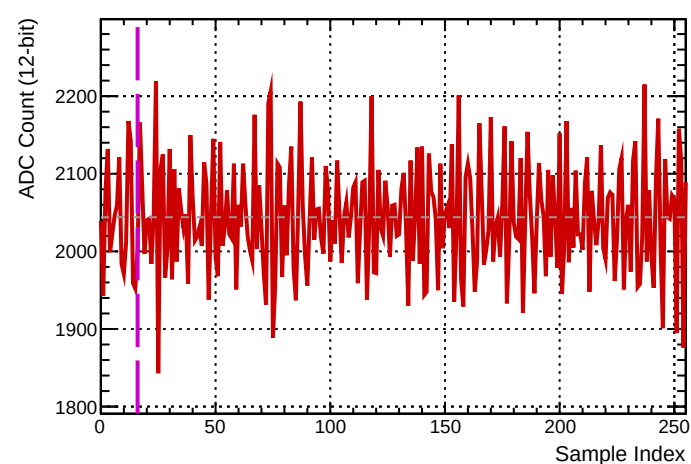
Ch 3 | Event #48 | Label: **SIGNAL** | CNN Score: 3500



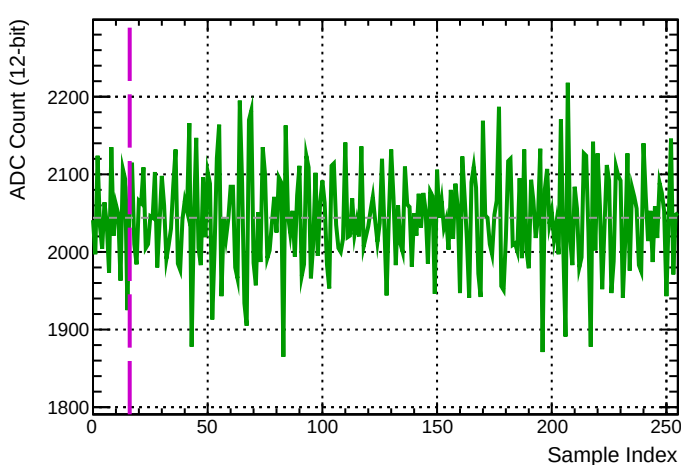
Ch 0 | Event #862 | Label: Background | CNN Score: 6743



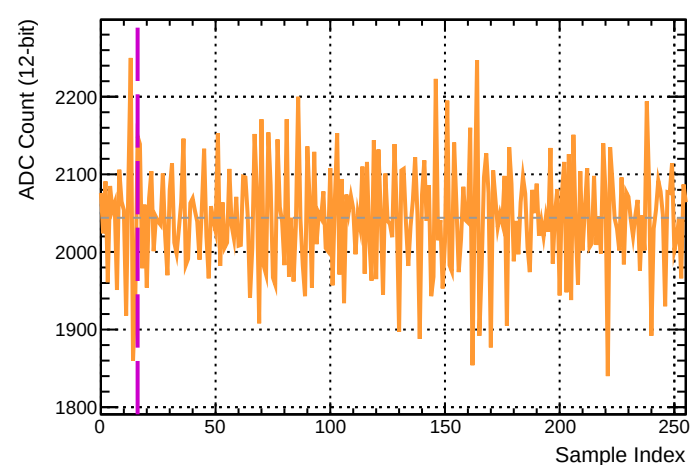
Ch 1 | Event #862 | Label: Background | CNN Score: 6743



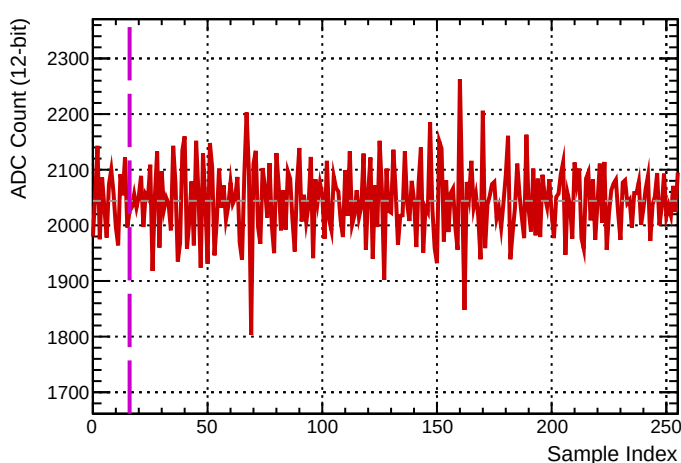
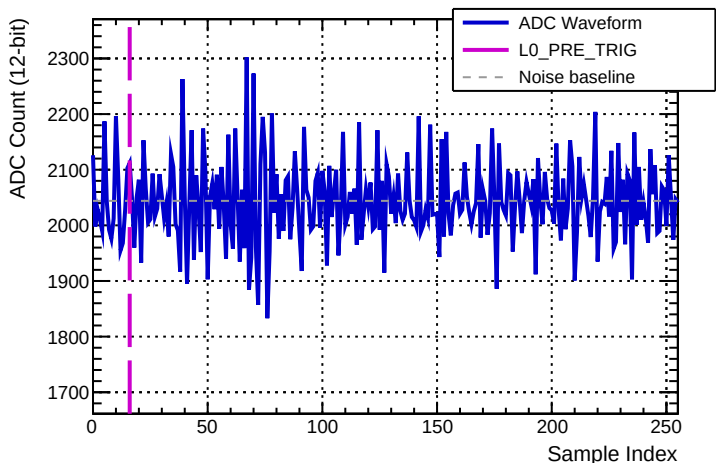
Ch 2 | Event #862 | Label: Background | CNN Score: 6743



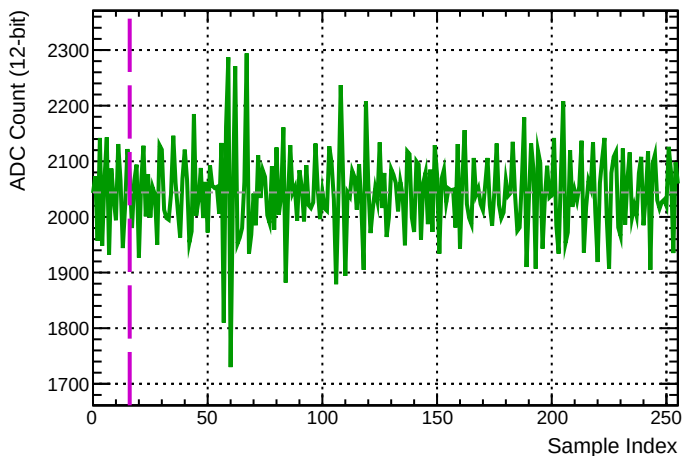
Ch 3 | Event #862 | Label: Background | CNN Score: 6743



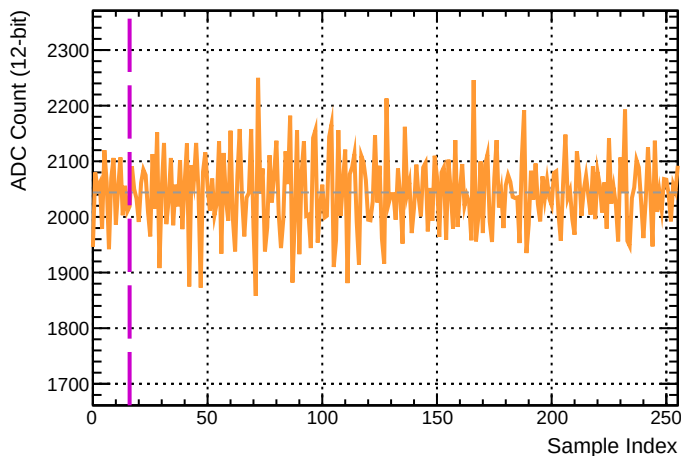
Ch 0 | Event #50 | Label: **SIGNAL** | CNN Score: 2529



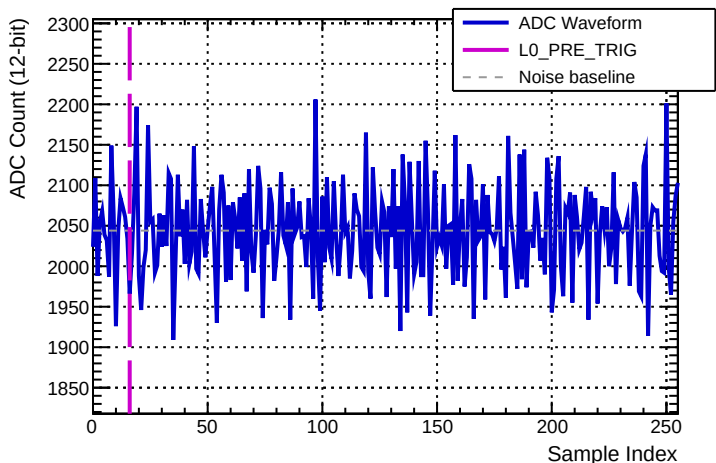
Ch 2 | Event #50 | Label: **SIGNAL** | CNN Score: 2529



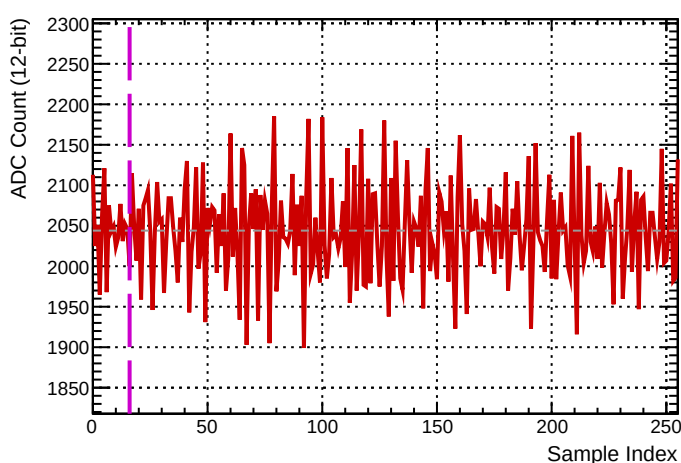
Ch 3 | Event #50 | Label: **SIGNAL** | CNN Score: 2529



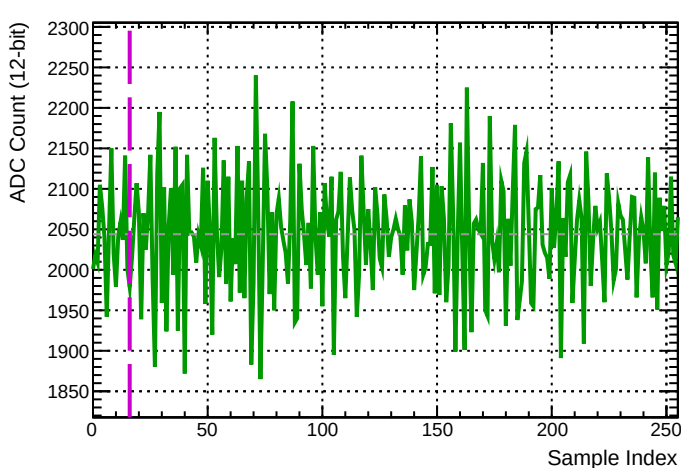
Ch 0 | Event #855 | Label: Background | CNN Score: 123801



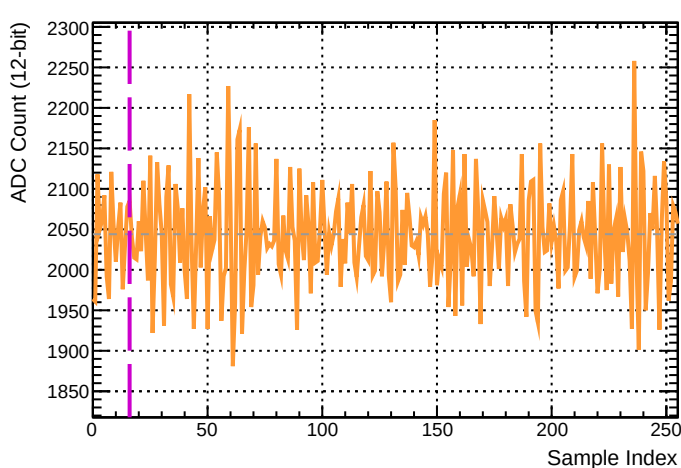
Ch 1 | Event #855 | Label: Background | CNN Score: 123801



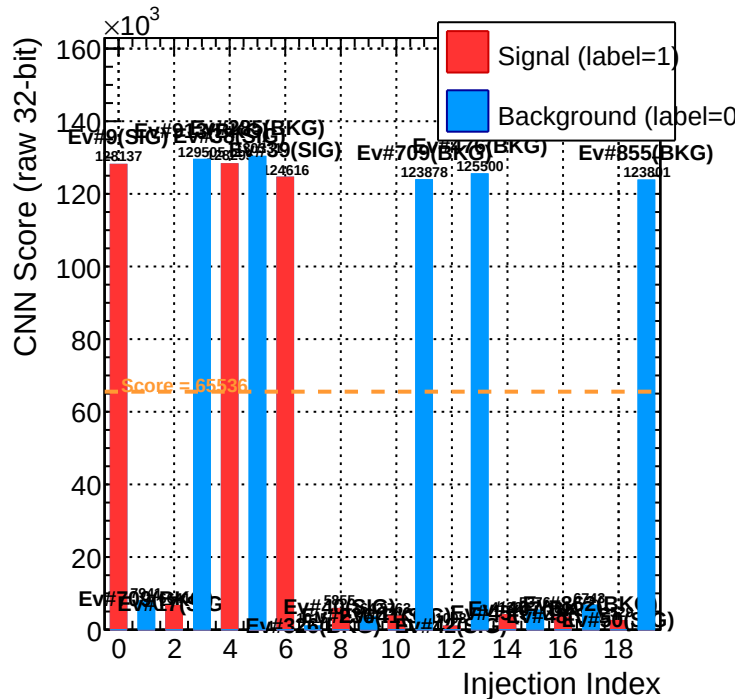
Ch 2 | Event #855 | Label: Background | CNN Score: 123801



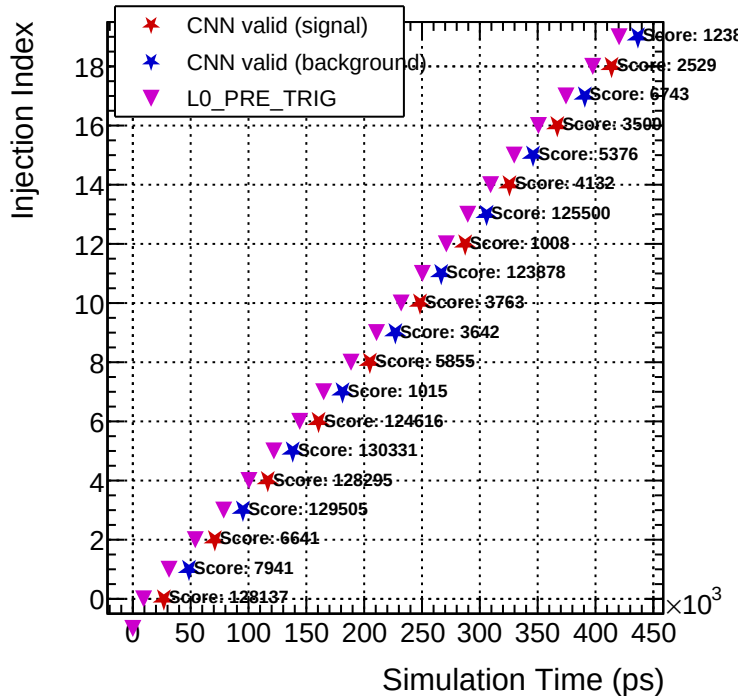
Ch 3 | Event #855 | Label: Background | CNN Score: 123801



CNN Output Score per Injected Event



Event Timeline



Simulation Results Summary

Test	EvIdx	Label	CNN Score	#Triggers	Result
0	#9	SIGNAL	128137 (0x1F489)	1	Correct
1	#709	Background	7941 (0x1F05)	1	Wrong
2	#17	SIGNAL	6641 (0x19F1)	1	Correct
3	#913	Background	129505 (0x1F9E1)	1	Wrong
4	#38	SIGNAL	128295 (0x1F527)	1	Correct
5	#285	Background	130331 (0x1FD1B)	1	Wrong
6	#39	SIGNAL	124616 (0x1E6C8)	1	Correct
7	#326	Background	1015 (0x3F7)	1	Wrong

